Anirudh Malik

ML Engineer — Data Scientist

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LinkedIn GitHub Portfolio Delhi, India

Summary

I am a Scientific Software Engineer and Data Scientist with a robust background in AI product development, MLOps, and cloud deployment. I possess strong skills in Python, SQL, and C++, as well as familiarity with frameworks like TensorFlow and PyTorch. I am committed to building scalable ML pipelines and solving complex real-world problems in science and computation.

Experience

AI/ML Specialist — ***Imagenators***
Noida — **05/2025 – 07/2025**

- Designed and deployed a containerized RAG system using FastAPI, LangChain, FAISS, and Gemini 1.5 Flash
- Developed endpoints to ingest, chunk, embed, and query PDF documents with real-time LLM responses
- Ensured production readiness with metadata storage in PostgreSQL, Docker Compose orchestration, and automated testing
- Built a RAG-based alumni chatbot using Ollama + Mistral, LangChain, Django, and ChromaDB
- Benchmarked GenAI vs classical NLP for accurate information retrieval

Trainee Developer — ***Dev Group***
New Delhi — **02/2025 – 05/2025**

- Prototyped AI tools using Python, REST APIs, and LangChain
- Developed a CO_2 emission estimator using Flask, Pickle, MLFlow on AWS

Research Associate — **IIT Delhi**
Delhi, India — **04/2025 – Present**

- Simulated Higgs boson production via $gg \to H \to \tau^+\tau^-$ using `MadGraph5`, `Pythia8`, and `Delphes` for detector effects
- Performed kinematic reconstruction and applied optimized selection cuts in `ROOT` and Python to enhance signal-to-background ratio
- Conducted exploratory data analysis (EDA) on CMS MiniAOD datasets to compare reconstructed vs. generator-level τ leptons

Graduate Researcher — **University of Sheffield, UK**
Sheffield, UK — **09/2023 – 09/2024**

- Built PICNN classifiers for $Z \to e^+e^-$ using ATLAS MC data
- Applied Optimal Transport to match AF3 and Geant4 simulations

Undergraduate Researcher — **University of Delhi**
Delhi, India — **09/2022 – 05/2023**

- Simulated surface plasmon resonance (SPR) via DDSCAT
- Studied dielectric, piezoelectric behavior of $AgNO_3$ nanomaterials

Education

- **MSc in Particle Physics**, University of Sheffield, UK
  09/2023 – 09/2024 — *First Division*
- **BSc in Physical Science**, University of Delhi, India
  09/2020 – 09/2023 — *Distinction*
- **Senior Secondary (ISC)**, Mount Carmel School
  09/2018 – 09/2020 — *94%*

Skills

- **Languages:** Python, SQL, C++, Bash
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn
- **MLOps:** MLflow, Docker, Airflow, Kubernetes
- **APIs/Backends:** REST, FastAPI, Flask, Django
- **LLM/RAG:** LangChain, Ollama, ChromaDB, Gemini 1.5
- **Data:** Pandas, NumPy, PostgreSQL, MongoDB
- **Visualization:** Matplotlib, Seaborn
- **Cloud:** AWS (EC2, Lambda, S3)
- **Tools:** Git, GitHub, GitLab, Pickle
- **Physics/Sim:** ROOT, Geant4, MadGraph5, Pythia8
- **Big Data:** Apache Spark, Hadoop
- **NLP:** Transformers, HuggingFace

Key Achievements

- **ATLAS Electron Identification**
  Developed PICNN for $Z \to e^+e^-$ classification using ATLAS MC data; improved data-MC agreement with Optimal Transport-based domain adaptation.
- **High-Energy Physics Research**
  Simulated $gg \to H \to \tau^+\tau^-$ events using MadGraph5 and Pythia8; optimized signal reconstruction using ROOT and CMS MiniAOD datasets.
- **Academic Distinction**
  Graduated with First Division (MSc) and Distinction (BSc) in Physics from globally recognized institutions.

Projects

3D Schrödinger Solver
- Solved 3D time-independent Schrödinger equation and benchmarked CPU vs GPU

Fraud Detection System
- Spark-based fraud detection system using AWS Lambda

YouTube Comment Analysis
- Built a transformer-based sentiment classifier

Certifications

- **Data Science, ML, DL, NLP**
- **Mastering SQL & Analytics**

Languages

English — Professional **Hindi** — Native